star must implode. In 1939, J. Robert Oppenheimer and his student Hartland Snyder used Einstein's laws to discover that, if the implosion is precisely spherical, the imploding star *must* create a black hole around itself, and then create a singularity at the hole's center, and then get swallowed into the singularity. No matter is left behind. None whatsoever. The resulting black hole is made entirely from warped space and time. Over the decades since 1939, physicists using Einstein's laws have shown that if the imploding star is deformed and spinning, it will also produce a black hole. Computer simulations reveal the full details.

Astronomers have seen compelling evidence for many black holes in our universe. The most beautiful example is a massive black hole at the center of our Milky Way galaxy. Andrea Ghez of UCLA, with a small group of astronomers that she leads, has monitored the motions of stars around that black hole (Figure 5.8). Along each orbit, the dots are the star's position at times separated by one year. I marked the black hole's location by a white, five-pointed symbol. From the stars' observed motions, Ghez has deduced the strength of the hole's gravity. Its gravitational pull, at a fixed distance, is 4.1 million times greater than the Sun's pull at that distance. This means the black hole's mass is 4.1 million times greater than the Sun's!



Fig 5.7. Black-hole scientists. *Left to right*: Karl Schwarzschild (1873–1916), Roy Kerr (1934–), Stephen W. Hawking (1942–), J. Robert Oppenheimer (1904–1967), and Andrea Ghez (1965–).